THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is an example of a gas in liquid solution used in the fire extinguisher? 1

(a) oxygen in oil **(b)** nitrogen in air
(c) helium in water **(d)** carbon dioxide in water

2. Assertion (A): Muddy water can be cleared by filtration. Reason (R): Muddy water is a suspension. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. The solubility of a gas in water at normal pressure is x. At a much higher pressure and the same temperature it will be: 1

(a) greater than x **(b)** equal to x **(c)** less than x **(d)** zero

4. Camphor dissolves in alcohol but sulphur does not. This shows: 1

(a) alcohol is heavy **(b)** solubility depends on solute and solvent
(c) sulphur is a gas **(d)** camphor is a metal

5. Assertion (A): Dilution of a solution decreases its mass percentage. Reason (R): The mass of the solute increases. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A student dissolved 10 g of salt in 90 g of water, then 10 g more. The second 10 g dissolves fully. The solution after the second addition has mass percentage: 1

(a) about 18 per cent **(b)** 10 per cent
(c) 20 per cent **(d)** 1 per cent

7. A sealed bottle of warm soda opens with a sudden spray. The reason is: 1

(a) high gas pressure from reduced gas solubility
(b) more oxygen
(c) less sugar
(d) more salt

8. Assertion (A): Oil does not mix with water. Reason (R): Oil molecules are unlike water molecules in nature. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. A saturated solution of salt is prepared at 25 degrees Celsius and kept in a closed bottle. A day later some salt crystals are at the bottom. This shows: 1

(a) equilibrium between dissolved and undissolved salt
(b) salt reacts with the bottle
(c) temperature rises
(d) water disappears

10. To separate a mixture of two miscible liquids with different boiling points one should use: 1

(a) distillation **(b)** filtration **(c)** evaporation **(d)** decantation

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a hot saturated solution give crystals on cooling but a dilute solution does not? 2

12. Why are fish found in greater numbers in cold waters? 2

13. State the difference between 'dissolve' in a physical sense and 'react' in a chemical sense with two examples. 2

14. Why are nasal sprays and eye drops made from sterile, dilute solutions? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why sea salt farming (salt pans) is more efficient in summer. 3

16. What are the ways of increasing the concentration of a solution? Give three. 3

17. Explain the role of solvents in paint, perfume and medicines. 3

OR

Why is it easier to remove a stain with a suitable solvent than with plain water? Give two examples. 3

18. The concentration of a solution of copper sulphate is judged by colour. How? 3

OR

Why does sugar syrup become thick and sticky on boiling? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how the dissolved gases and temperature in water bodies affect aquatic life, including the effect of summer, pollution and eutrophication. 5

OR

Explain, with examples, how the solubility of solids changes with temperature and how crystals are formed from solutions, applying these ideas to sugar candy or rock sugar. 5

20. Calculate the masses: 5 per cent salt solution (200 g) and 15 per cent salt solution (300 g) are mixed. Find the percentage of the mixture and the water in it. 5

OR

A student measures the mass of a beaker with 100 g water, adds 20 g of salt and stirs till it dissolves. Predict the final mass, the mass percentage, and what happens to the mass if the water is then evaporated completely. 5

END OF QUESTION PAPER — Total: 40 Marks